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Electric Network Topology Optimization with Distributed Generation using Pandapower Library

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ABSTRACT

The paper describes the selection of the optimal topology for a distributed electric network with renewable energy sources. The Newton-Raphson method is used to calculate power flows, which is the basis of the Pandapower library. The library is a Python-based tool for modeling, analyzing, and optimizing electrical networks. It allows for the creation and modification of electrical schematics, calculation of power flows, determination of voltages and currents, as well as optimization of system performance. The result of the work is the determination of the optimal configuration in terms of minimizing electrical energy losses in network lines. The developed software allows for loading data on network nodes and lines, including active and reactive power consumption and generation, performing steady-state calculations with determination of losses in lines. Due to the automation of calculations, it is possible to perform calculations for all permissible network topologies and select the optimal one for the given conditions. The result of the work is the determination of the optimal configuration in terms of minimizing electrical energy losses in network lines.

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1. INTRODUCTION

With the development of energy, the possibilities of extracting electricity are increasing. New technologies are used to reduce the harmful effects on the environment, increase efficiency, and reduce losses in the transmission of electricity. One of such innovations includes systems with distributed generation based on renewable energy sources. The principle of distributed generation consists in the close proximity of power sources to consumers, that is, relatively small long lines transmitting electricity [1]–[4]. An important task in the construction of such lines is to choose the optimal network topology at a certain load. Different optimization methods are used, both deterministic and heuristic and metaheuristic [5]–[7]. If the electric power system contains renewable energy sources, the high efficiency of their generation leads to the need for frequent change of the network topology [5], [8]–[10].

To calculate line losses, there are many software development tools that include various libraries. Each network has its own properties: the number of nodes, the number of generation sources, the number of external sources, the load, the number and location of switching devices (switches, disconnectors), line parameters (line length, linear active resistance, linear inductive resistance, capacitance, maximum permissible current on the line, etc.).

2. METHODOLOGY

The study of a distributed generation system was performed using the Newton-Raphson method [11], [12].

The idea of the method is to linearize, replacing a nonlinear equation with a linear one to solve the equation $f(x) = 0$. The equation is reduced to the form $x = \varphi(x)$.

There are many nonlinear equations:

$$f_i(x_1, x_2, \dots, x_n) = y_i, i = 1, 2, \dots, n \tag{2.2}$$

With a set of initial approximations: $x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)}$, we denote the set of corrections $\Delta x_1^{(0)}, \Delta x_2^{(0)}, \dots, \Delta x_n^{(0)}$. Then, using Taylor series and neglecting higher-order terms, we obtain a corrected system of equations

$$\left(x_1^{(0)} + \Delta x_1, x_2^{(0)} + \Delta x_2, \dots, x_n^{(0)} + \Delta x_n\right) = y_i, \tag{2.3}$$

where Δx_i are the corrections for $x_i, i = 1,2, \dots, n$.

A set of linear equations that define the tangent hyperplane to the function $f_i(x)$ at a given iteration point $x_i^{(0)}$, is obtained as

$$\Delta Y = J * \Delta X, \tag{2.4}$$

где ΔY – the column vector defined by $y_i - f_i(x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)})$, ΔX – vector column of correction terms Δx_i, and J – the Jacobi matrix for the function f, it has the form of the first order of partial derivatives, defines $x_i^{(0)}$. The corrected solution is obtained in the form:

$$x_i^{(1)} = x_i^{(0)} + \Delta x_i \tag{2.5}$$

The Jacobi matrix is defined by

$$J_{ik} = \frac{\partial f_i}{\partial x_k} \tag{2.6}$$

The above Newton-Raphson method is used to solve the problem of power flows. The matrix J – is a sparse matrix and is particularly suitable for calculating flow distribution. Triangular decomposition and reverse substitution methods provide fast and efficient convergence to a flow distribution solution. This method has quadratic convergence and therefore converges very quickly when the solution point is close.

All the data necessary for calculations on nodes, lines, consumption and generation schedules are tabular data that can be easily loaded into software by reading XLS files through the Pandas library of the Python programming language [13], [14]. Then, based on the uploaded data, a network model is created using the Pandapower library. It is also used to calculate the steady-state mode of the network and flow distribution using the Newton-Raphson method. Due to the changing nature of generation and consumption, for each hour of the day, it is possible to determine its optimal network topology in terms of power losses.

The encoding of the topology is performed as follows. The solution is encoded by a list of *N* values, where *N* is the number of all power lines in the network (let's take 20 lines as an example). The list items take the values 0 (the line is disabled) or 1 (enabled). The total number of possible combinations is 220 = 1048576. At the same time, it is obvious that most of the solutions do not require mode calculation, since they either contain rings, or do not provide a balance of generation and consumption if any part of the network turns out to be isolated and its consumption is lower than its own generation.they contain rings, or do not provide a balance of generation and consumption if any part of the network turns out to be isolated and its consumption is lower than its own generation.

The algorithm for finding the optimal configuration can be written as follows:

1. The cycle of generating topologies – binary codes from [0, 0, …, 0] to [1, 1, … 1].
   1.1. If the topology contains a ring, proceed to the next iteration of the loop.
   1.2. If the topology does not provide a balance of consumption and generation, proceed to the next iteration of the cycle.
   1.3. Calculation of flow distribution and active power losses in network lines.
   1.4. If the losses are lower than with the best topology found earlier, take the current topology as the best.

1.5. Go to the next iteration of the loop.

The best topology found in cycle 1 is the desired solution.

The algorithm must be executed for each moment in time. In the calculated examples, the time step will be 1 hour. Due to the preliminary cutting off of obviously inoperable and inefficient topologies, the algorithm is able to calculate the optimal topology on a personal computer not in real time, but with sufficient speed to perform calculations in increments of 1 hour.

Depending on the mode of operation of renewable energy sources, the flow of power in the network may change its direction, unlike centralized electric power systems, where the flow of power is predominantly unidirectional. As a result, the current network topology may not be optimal in terms of power losses in power transmission lines and their insufficient/excessive load.

The radial configuration of a 15-node electric power system with two solar power plants with an installed capacity of 1.8 MW and 2.5 MW is considered. The rated voltage of each bus is 10 kV. The network diagram is shown in Figure 1. The normally switched on lines are shown in black, and the normally disconnected lines are shown in red. Each line has its own parameters.

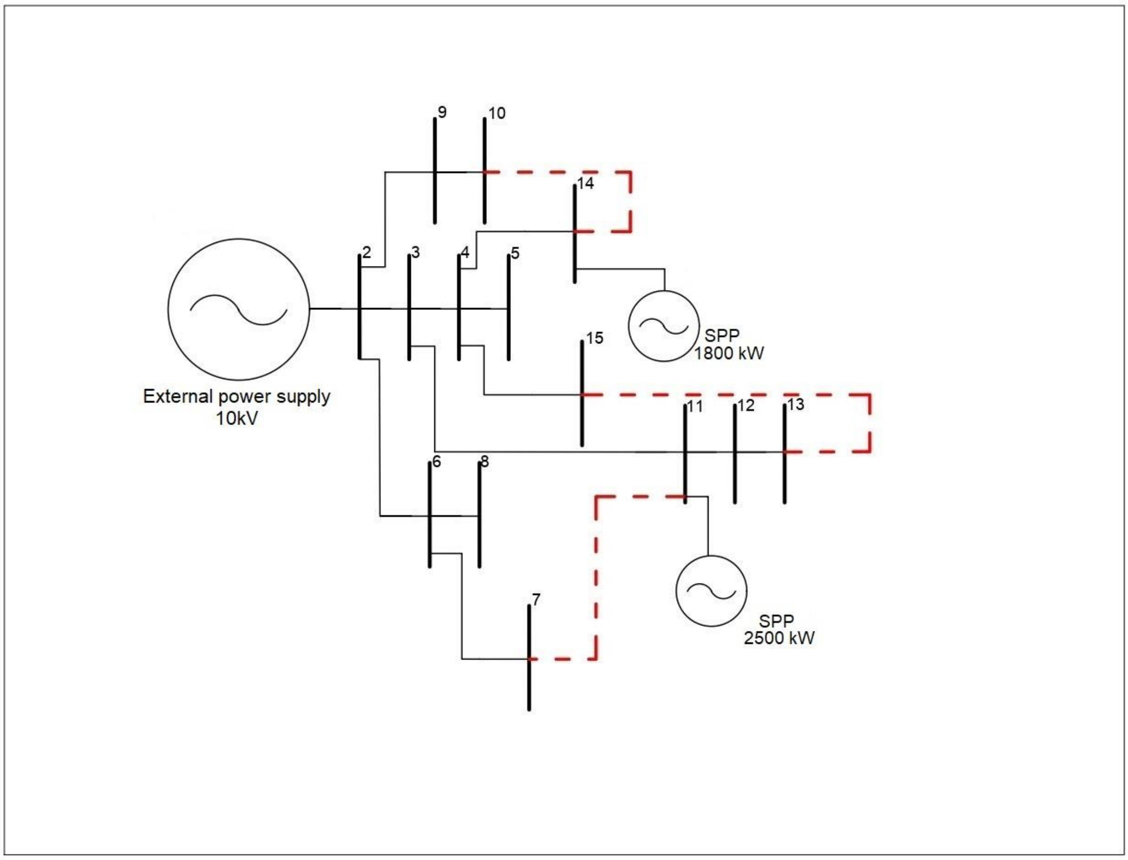

Figure 1. The structure of the electrical network

To model the network, the parameters of electrical loads are used, which correspond to their line.

Then you need to connect to the Python library – "Pandapower".

Pandapower is a library whose function is to model the parameters of an electrical network. The Pandapower electrical network consists of tables for each type of network element. For example, a separate table for transformers, a separate table for power lines, etc. Each grid element table consists of a column for each parameter and a row for each network element. Various network elements can be generated using the program code.

3. RESULTS AND DISCUSSION

The algorithm must be executed for each moment in time. In the calculated examples, the time step will be 1 hour (Figure 2 shows output). Due to the preliminary cutting off of obviously inoperable and inefficient topologies, the algorithm is able to calculate the optimal topology on a personal computer not in real time, but with sufficient speed to perform calculations in increments of 1 hour.

```
Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
Collecting pandapower
  Downloading pandapower-2.9.0.zip (6.0 MB)
     |████████████████████████████████| 6.0 MB 5.0 MB/s
Requirement already satisfied: pandas>=0.17 in /usr/local/lib/python3.7/dist-packages (from pandapower) (1.3.5)
Requirement already satisfied: networkx>=2.5 in /usr/local/lib/python3.7/dist-packages (from pandapower) (2.6.3)
Requirement already satisfied: scipy in /usr/local/lib/python3.7/dist-packages (from pandapower) (1.4.1)
Requirement already satisfied: numpy>=0.11 in /usr/local/lib/python3.7/dist-packages (from pandapower) (1.21.6)
Requirement already satisfied: packaging in /usr/local/lib/python3.7/dist-packages (from pandapower) (21.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.7/dist-packages (from pandapower) (4.64.0)
Requirement already satisfied: python-dateutil>=2.7.3 in /usr/local/lib/python3.7/dist-packages (from pandas>=0.17->pandapower) (2.8.2)
Requirement already satisfied: pytz>=2017.3 in /usr/local/lib/python3.7/dist-packages (from pandas>=0.17->pandapower) (2022.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.7.3->pandas>=0.17->pandapower) (1.15.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in /usr/local/lib/python3.7/dist-packages (from packaging->pandapower) (3.0.9)
Building wheels for collected packages: pandapower
  Building wheel for pandapower (setup.py) ... done
  Created wheel for pandapower: filename=pandapower-2.9.0-py3-none-any.whl size=5931108 sha256=5ad506fe33dc39f850cdc9d892e47ed09a1487a9d87ae609ea8fabedf4fb261a
  Stored in directory: /root/.cache/pip/wheels/94/44/e3/be45d434be4b2427df371bf379756058ae2485e721c8acca2a
Successfully built pandapower
Installing collected packages: pandapower
Successfully installed pandapower-2.9.0
```

Figure 2. Output data informing about the download of the "Pandapower" library

The next step is to refer to the network parameters and determine the optimal circuit configuration. Having received data on the network load, we will determine by switching disconnectors which configuration has less power loss. After specifying the system and all the input data, we receive information about the built network (Figure 3).

```
This pandapower network includes the following parameter tables:
   - bus (15 elements)
   - load (14 elements)
   - sgen (2 elements)
   - switch (3 elements)
   - ext_grid (1 element)
   - line (17 elements)
 and the following results tables:
   - res_bus (15 elements)
   - res_line (17 elements)
   - res_ext_grid (1 element)
   - res_load (14 elements)
   - res_sgen (2 elements)
```

Figure 3. Information about the network

Next, we get information about the most preferred topology with the least losses for each hour. By switching the disconnectors, we will determine which configuration has less power loss. For 0 hours, the least amount of losses is achieved when all disconnectors are in the closed state (Table 1 and Figure 4).

Table 1. Losses on the lines when all disconnectors are in the closed state

line	ql_mvar	pl_mw
1-2	0,7528	0,7361
2-3	0,1261	0,1228
3-4	0,0191	0,0186
4-5	0,0001	0,0000
2-9	0,0796	0,0522
9-10	0,0612	0,0411
2-6	0,1194	0,0805
6-7	0,0360	0,0243
6-8	0,0001	0,0000
3-11	0,0515	0,0348
11-12	0,0007	0,0005
12-13	0,0011	0,0007
4-14	0,0073	0,0049
4-15	0,0076	0,0031
10-14	0	0
13-15	0	0
7-11	0	0

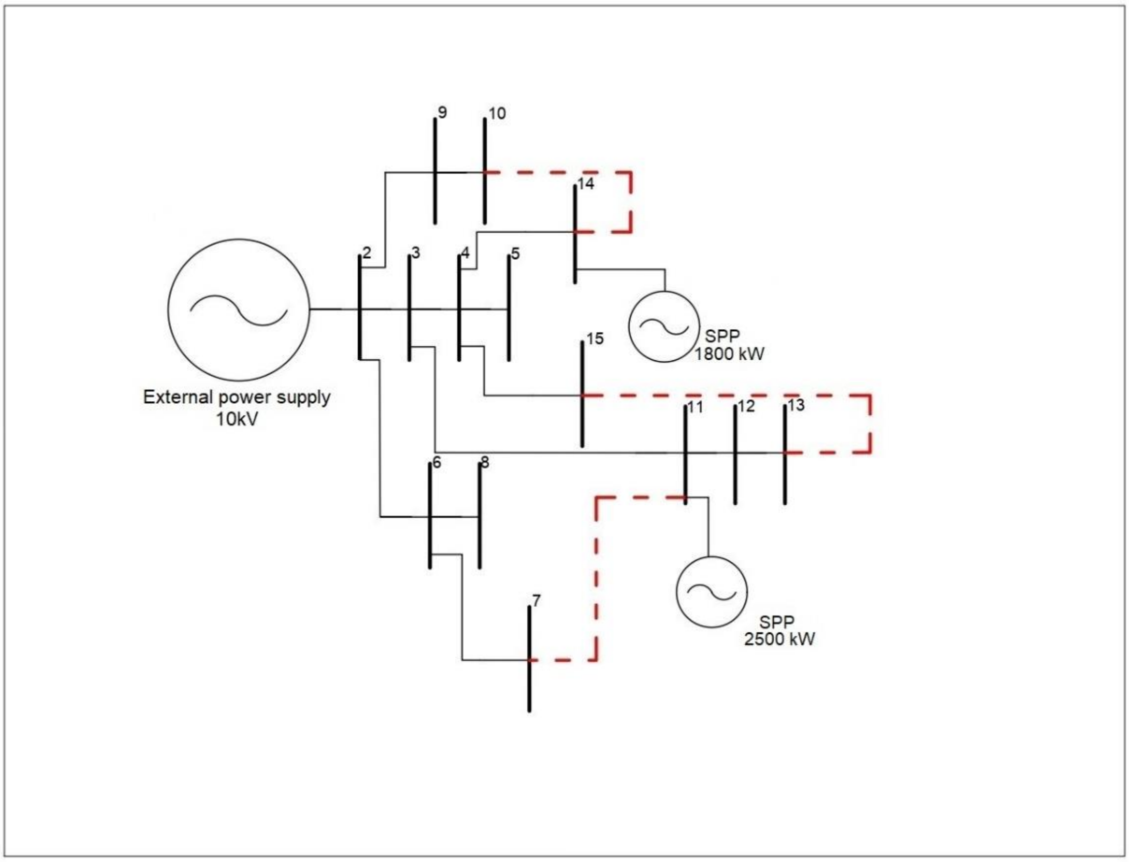

Figure 4. Optimal network configuration for 0 hours

Thus, the optimal topology for each hour is selected for each hour based on the daily load schedule and the parameters of the electrical network. When modeling the network, the same length of lines was specially set so that the active and inductive resistance had the same effect on flow distribution.

4. CONCLUSION

The developed software allows you to download data about nodes and lines of the network, including active and reactive load and generation capacities, to calculate the steady-state mode with the determination of losses in the lines, while due to automation of calculations it becomes possible to calculate the mode for all permissible network topologies and choose the optimal one for the specified conditions. Since the schedules of the load and generation of renewable energy are constantly changing, the approach used allows us to determine the optimal network configuration for the specified capacities of the load and generation of renewable energy.

The next steps of the study consist in the use of population-based optimization methods [15], [16] to solve problems of higher dimension and the transition to predictive optimization on the basis of forecasting the generation from renewable energy sources [17], [18].

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